
IEEE 802.11 Overview and Completed Amendments

Overview of the 802.11 Working Group

The IEEE 802.11 standard to date

Completed Amendments: Markets, use cases and key technologies

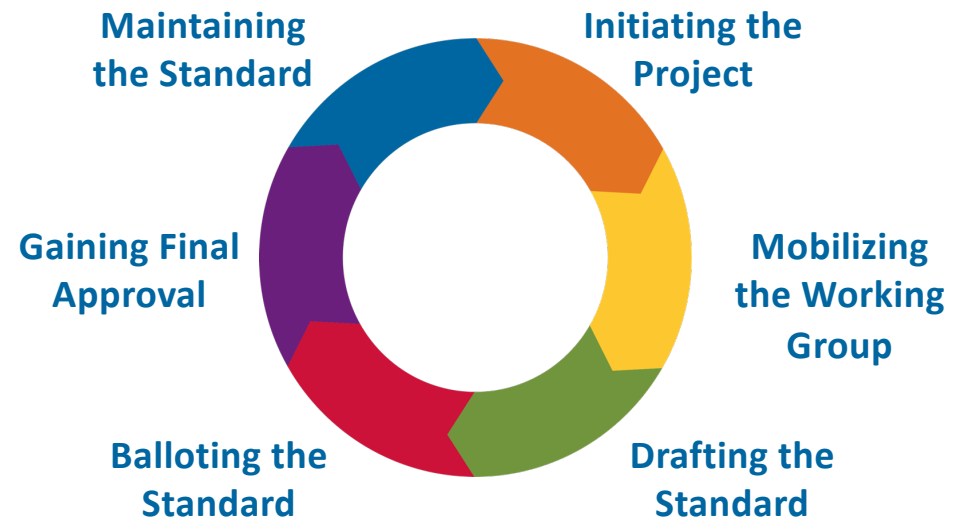
2025 January

Presenter:

“At lectures, symposia, seminars, or educational courses, an individual presenting information on IEEE standards shall make it clear that his or her views should be considered the personal views of that individual rather than the formal position, explanation, or interpretation of the IEEE.” IEEE-SA Standards Board Operation Manual (subclause 5.9.3)

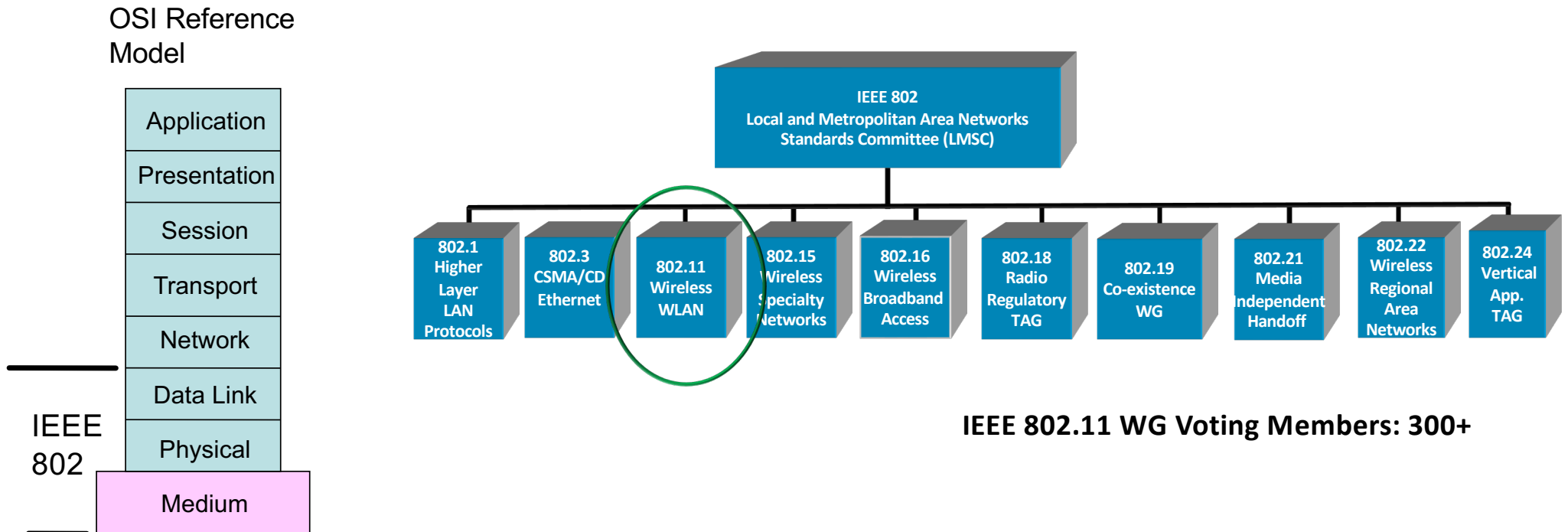
IEEE Standards Association

IEEE Standards drive the functionality, capability, and interoperability of a range of products and services that affect the way people live, work, and communicate.



The IEEE 802.11 Working Group is one of the most active WGs in 802

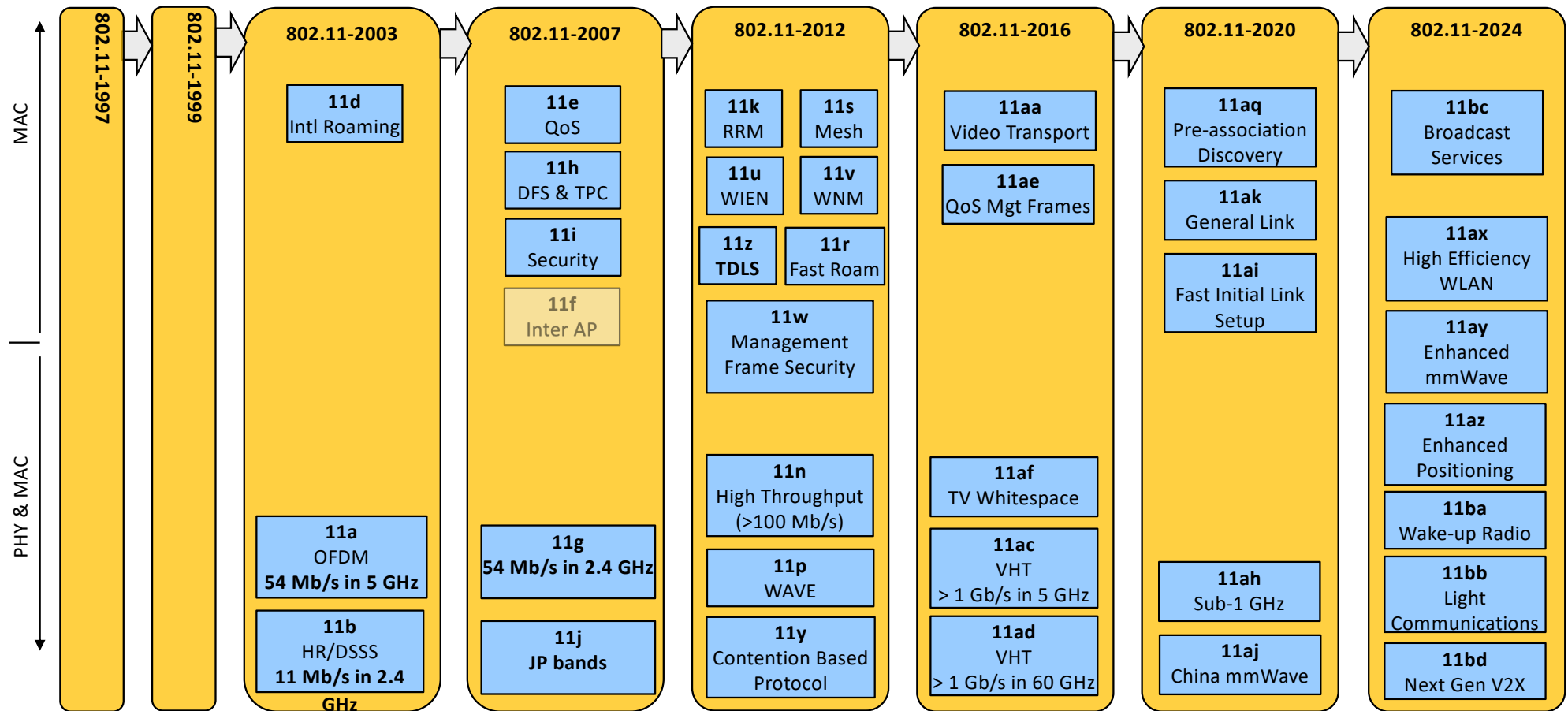
- Focus on **link and physical layers** of the network stack
- Leverage IETF protocols for upper layers



Type of Groups in 802.11

Type of Group	Description
WG	Working Group
SC	Standing Committee
TG	Task Group
SG	Study Group
TIG	Topic Interest Group
AHG	Ad Hoc Group

Development of the IEEE 802.11 Standard is ongoing since 1997



Completed 802.11 technologies to meet expanding market needs and leverage new technologies - 1

- 802.11ah – Operating in license-exempt bands below 1 GHz (excl. TV bands)
- 802.11ai – Fast initial link set-up
- 802.11aj – Enable 802.11ad operation in the Chinese 59-64 GHz frequency band.
- 802.11ak – General Link
- 802.11aq – Pre-association Service Discovery
- 802.11ax – Increased throughput in 2.4, 5 (and 6) GHz bands. Increased efficiency.
- 802.11ay – Support for 20 Gbps in 60 GHz band.
- ARC – Architecture (Ongoing discussion topics)

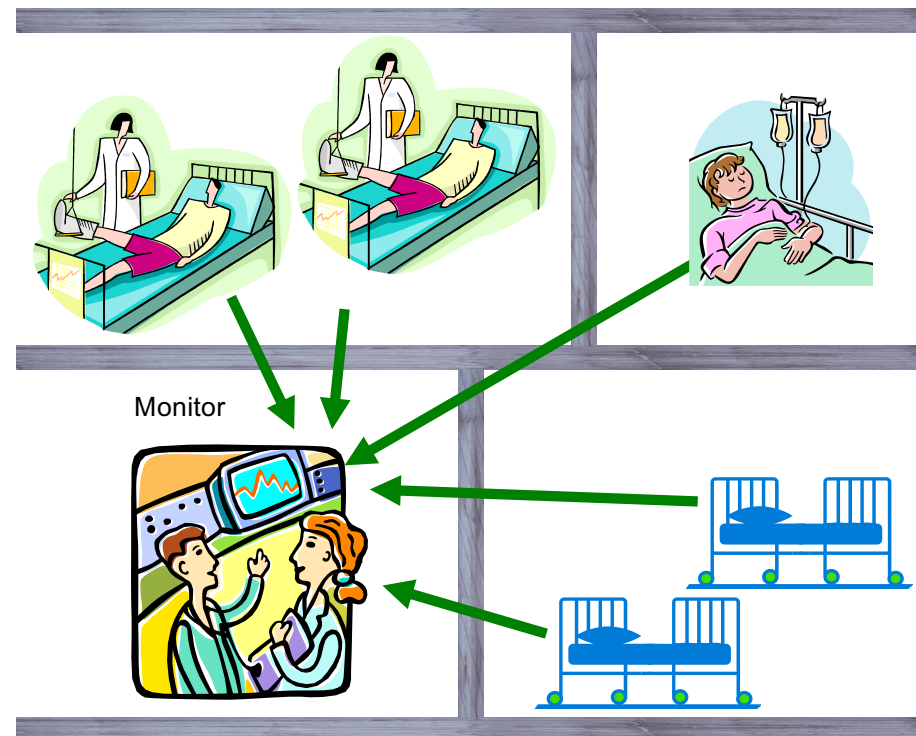
Completed 802.11 technologies to meet expanding market needs and leverage new technologies - 2

- 802.11az – 2nd generation positioning features.
- 802.11ba – Wake up radio. Low power IoT applications.
- 802.11bb – Light Communications
- 802.11bc – Enhanced Broadcast Service
- 802.11bd – Enhancements for Next Generation V2X
- 802.11be – Extremely High Throughput in 2.4, 5 and 6 GHz bands

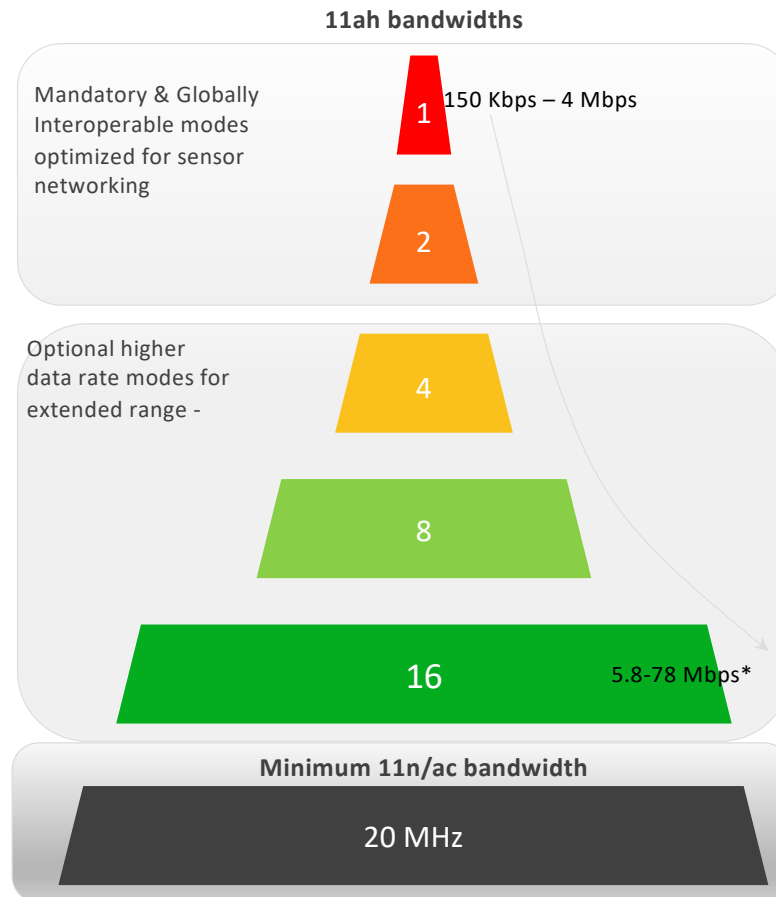


802.11ah defines OFDM PHY and MAC operating in license-exempt bands below 1 GHz (excl. TV bands)

- Define OFDM PHY and MAC operating in license-exempt bands below 1 GHz (excl. TV bands)
 - Extended range
 - Power efficiency
 - Large number of devices
- Potential Applications
 - Internet of everything (IoT)
 - Smart Grid
 - Healthcare
 - Smart Appliances
 - Wearable consumer electronics



802.11ah technical features



*For 1 spatial stream and normal guard interval only. 11ah supports up to 4 spatial streams and short guard interval providing data rates up to ~347 Mbps

PHY

- OFDM (FFT size 32 and 64)
- New reliable MCS-10 working with larger delay spread and Doppler for outdoor
- Diverse data rates: 150 Kbps-347 Mbps
- Range >1 km

MAC

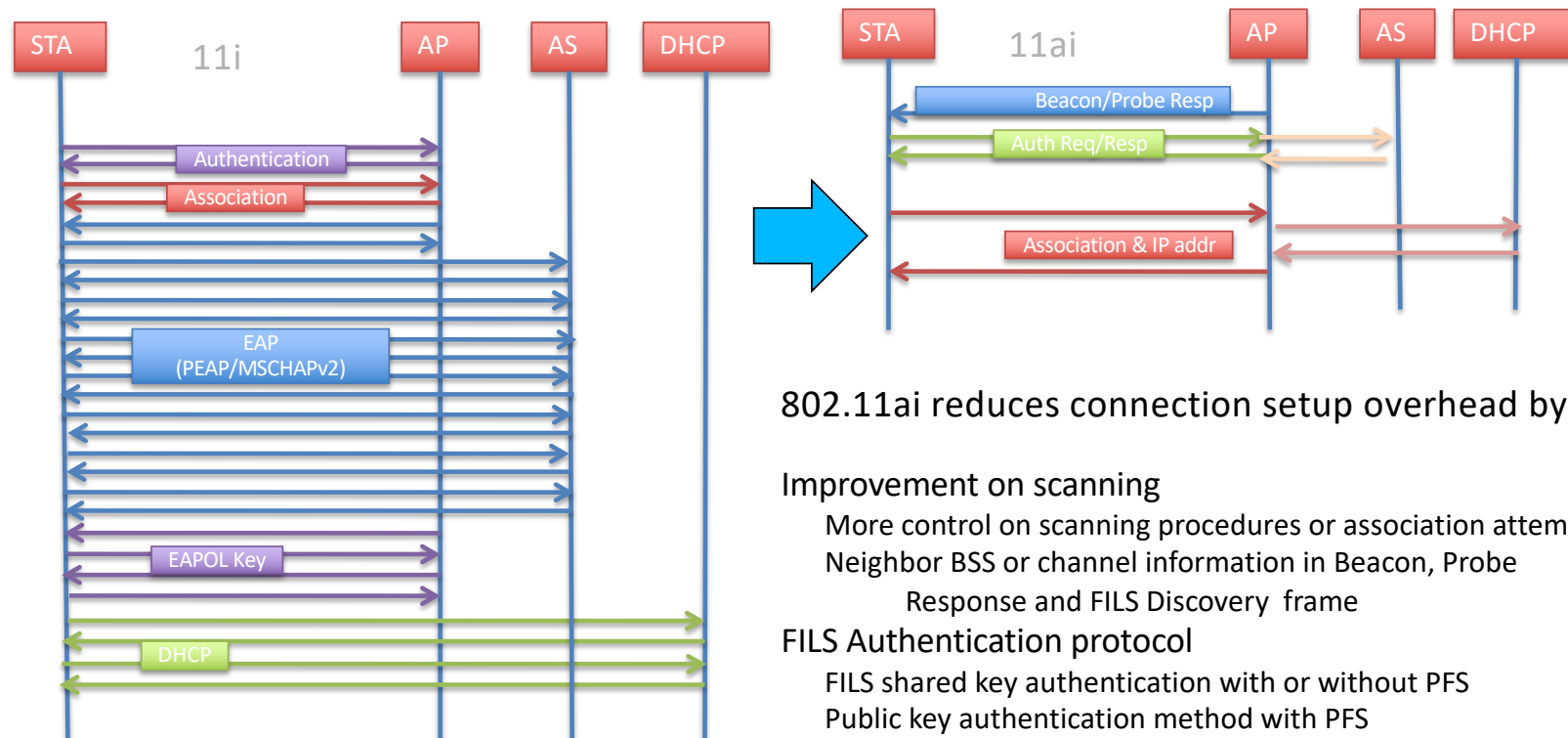
- Scalability up to 8191 devices per AP
 - ✓ *Restrict Access Window operation*
 - ✓ *Hierarchical TIM structure*
- Efficient frames and transmissions
 - ✓ *Short frame format*
 - ✓ *Short control/mgmt. frames*
 - ✓ *Asymmetric & bi-directional transmissions*
- Reducing power consumption
 - ✓ *Non-TIM operation*
 - ✓ *Target Wake Time mechanism*
 - ✓ *Extended sleeping and listen interval*
- Relay Operation

802.11ai defines fast initial link set-up methods

- This amendment defines mechanisms that provide IEEE 802.11 networks with fast initial link set-up methods which do not degrade the security currently offered by Robust Security Network Association (RSNA) already defined in IEEE 802.11.
- The project's primary need came from an environment where mobile users are constantly entering and leaving the coverage area of an existing extended service set (ESS).
 - scale with a high number of users simultaneously entering an ESS
 - minimize the time spent within the initial link set-up phase
 - securely provide initial authentication.



802.11ai technical features



802.11ai reduces connection setup overhead by:

Improvement on scanning

- More control on scanning procedures or association attempt
- Neighbor BSS or channel information in Beacon, Probe Response and FILS Discovery frame

FILS Authentication protocol

- FILS shared key authentication with or without PFS
- Public key authentication method with PFS

Higher Layer Setup during Association/Reassociation

- Higher Layer Protocol encapsulation
- FILS IP configuration

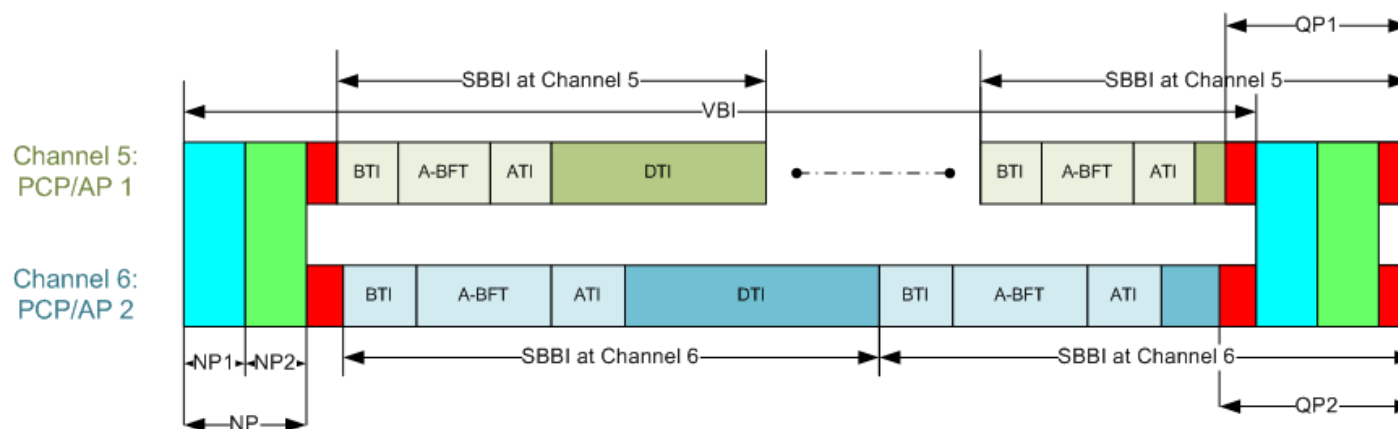
802.11aj defines modifications to 802.11ad to enable operation in the Chinese 59-64 GHz frequency band

- This amendment defines modifications to the IEEE P802.11ad Physical (PHY) layer and the Medium Access Control (MAC) layer to enable operation in the Chinese 59-64 GHz frequency band.
- The amendment maintains backward compatibility with 802.11ad when it operates in the 59-64 GHz frequency band.
- The amendment also defines modifications to the PHY and MAC layers to enable the operation in the Chinese 45 GHz frequency band. The amendment maintains the 802.11 user experience.

802.11aj technical features

11aj considers the following:

- Link Budget Analysis for 40-50 GHz Indoor Usage
- Multi-Carrier Training Field for OFDM Transmission in the 45GHz
- Packet Encoding Solution for 45GHz
- A Proposed frame structure is as follows



802.11ak enables an 802.11 connection to be used as a through link in a general network

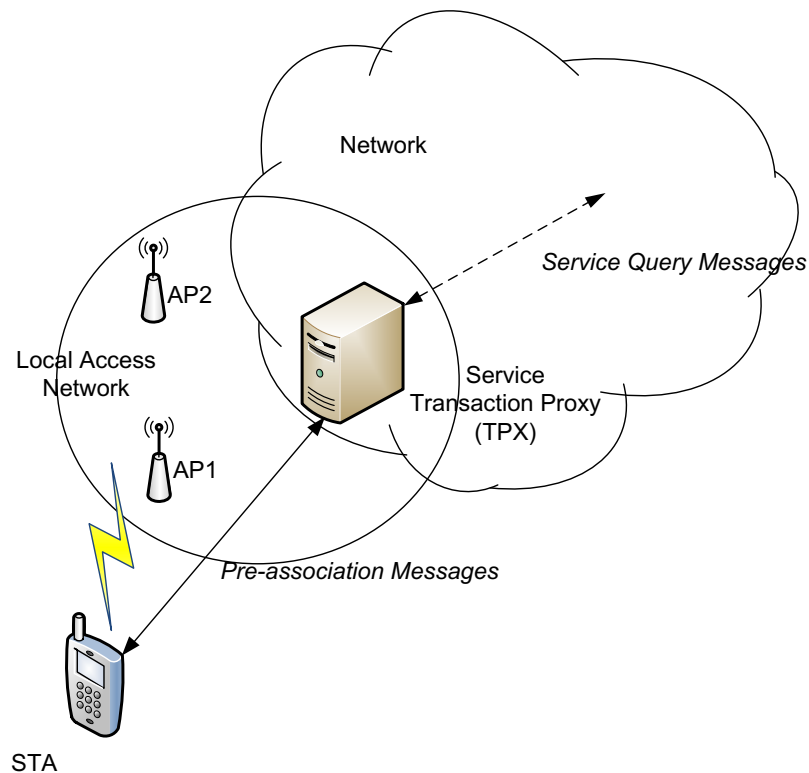
- This amendment enables an 802.11 connection to be used as a through link in a general network, not just as a connection to an end station at the edge of a network.
- Fully general mixed 802.11 and wired plug and play in the home.
- Data Center top-of-rack to top-of-rack connections for overflow traffic.
- Industrial and Enterprise network use.

802.11aq enables the delivery of pre-association Service Discovery information

- A typical use case is printer discovery in a hotel
- Pre-association protocol designed to discover services on a WLAN



802.11aq technical features



- Container MAC protocol to carry upper layer service discovery protocols (e.g. UPnP, Bonjour)
- Provisioning and configuration of services in the access point
 - Service Transaction Proxy is a logical element connected to the access point
- Universal identification of services
 - Using a hash name
 - Provide service attributes (e.g. 3D printer capability or point of sale service)
- Currently considering request/response or broadcast concept

802.11ax is focused on improving performance in dense environments

- Existing 802.11 WLAN systems serve dense deployments: 2019 Super bowl: 24TB* of data carried on WLAN network
- 802.11ax aims to further improve performance of WLAN deployments in dense scenarios
 - Targeting at least 4x improvement in the per-STA throughput compared to 802.11n and 802.11ac.
 - Improved efficiency through spatial (MU MIMO) and frequency (OFDMA) multiplexing.
- Dense scenarios are characterized by large number of access points and large number of associated STAs deployed in geographical limited region
 - e.g. a stadium or an airport.

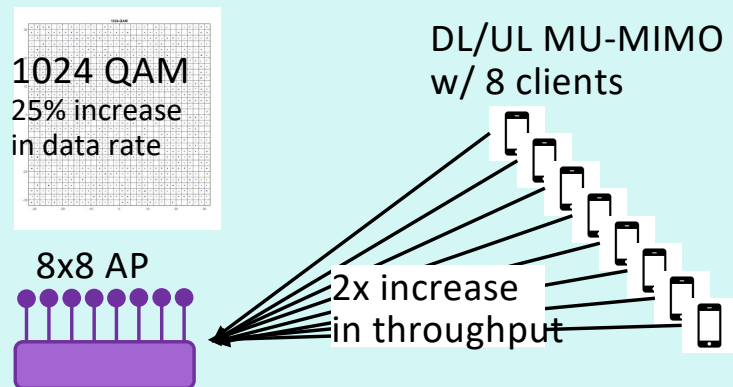


Access to Internet, latest airlines' announcements, and digital media such as movies and sport events

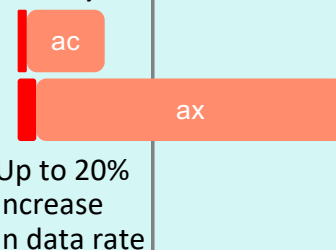
* <https://www.extremenetworks.com/resources/slideshare/wi-fi-engagements-from-super-bowl-iiii/>

802.11ax Categories of Enhancements

Spectral Efficiency & Area Throughput

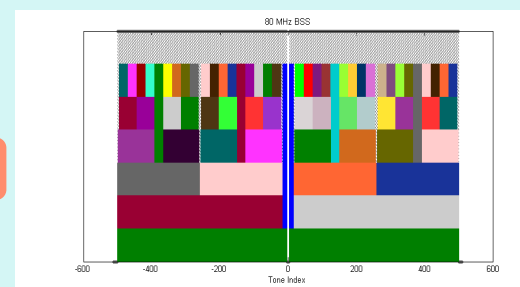


Long OFDM Symbol

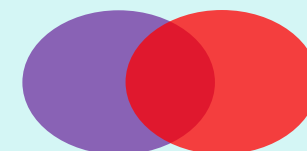


High Density

OFDMA

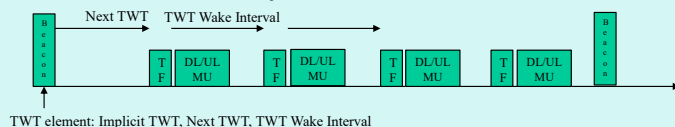


Spatial Reuse

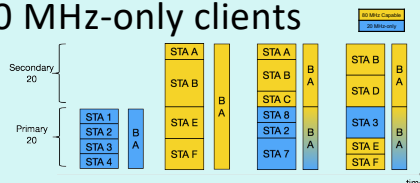


Power Saving

Scheduled sleep and wake times

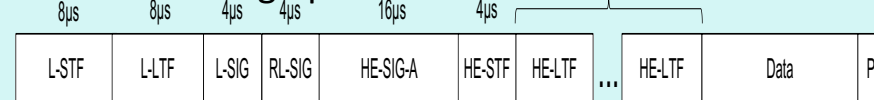


20 MHz-only clients

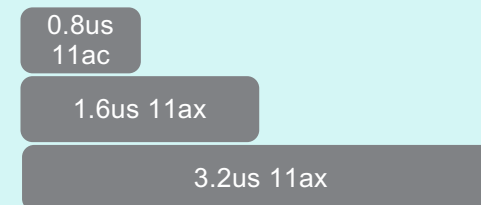


Outdoor / Longer range

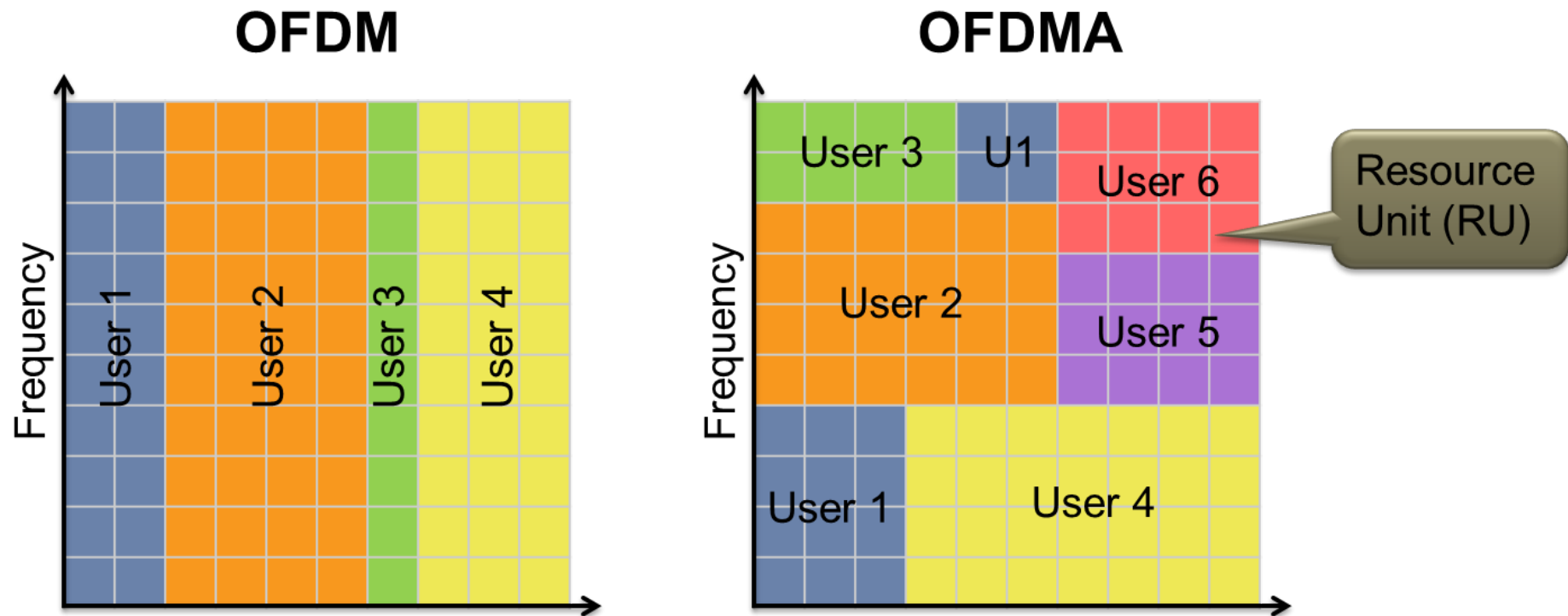
Extended range packet structure



Enhanced delay
spread protection-
long guard interval



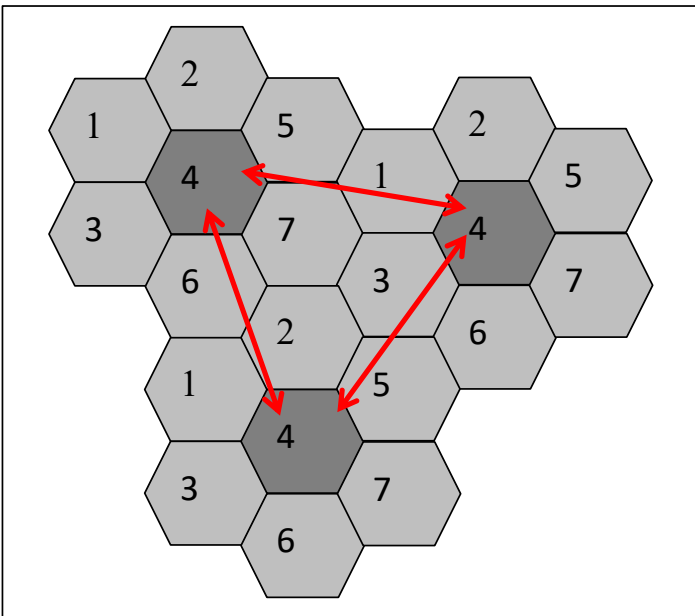
OFDMA enables further AP customization of channel use to match client and traffic demands



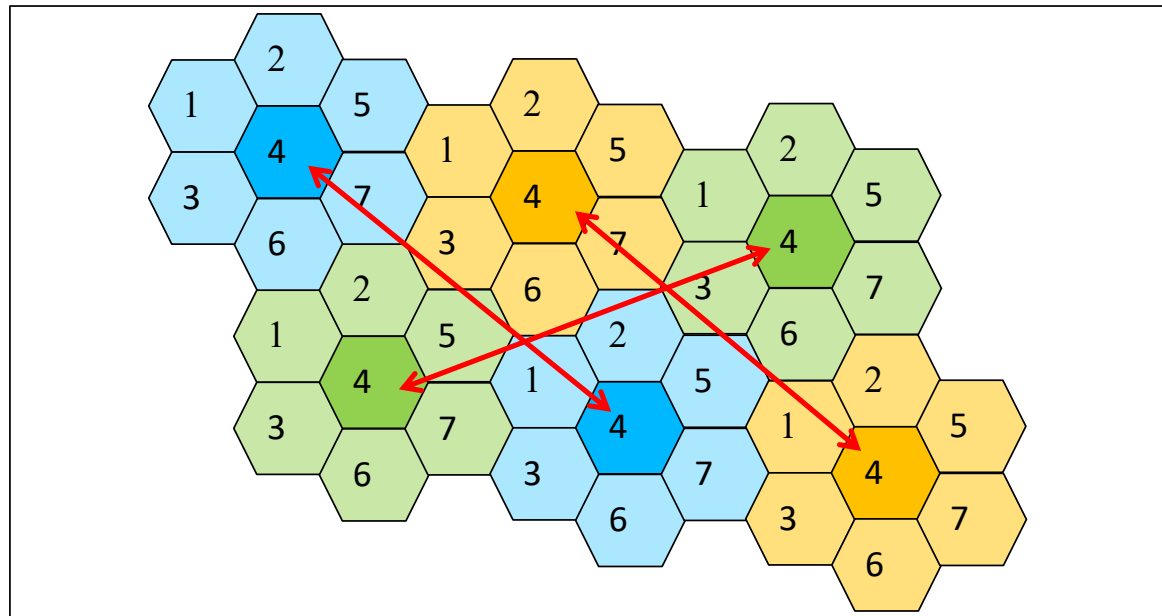
Increased efficiency for (high percentage of traffic) short data frames

BSS Coloring enables additional channel re-use

All same-channel BSS block

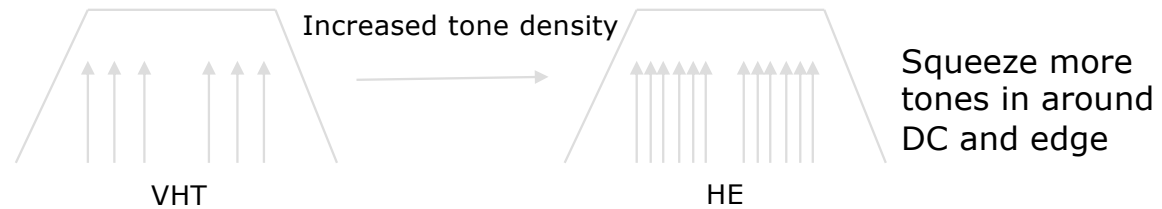


Same-channel BSS only block on Color Match

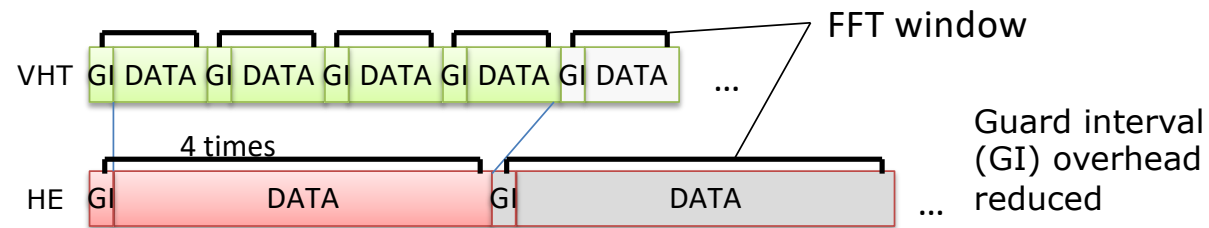


802.11ax Increases link efficiency

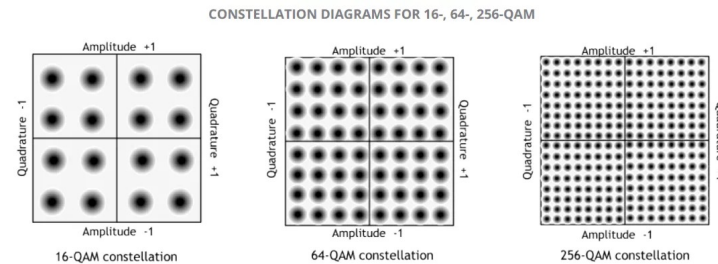
Frequency domain
(~5% gain)



Time domain
(~15% gain)

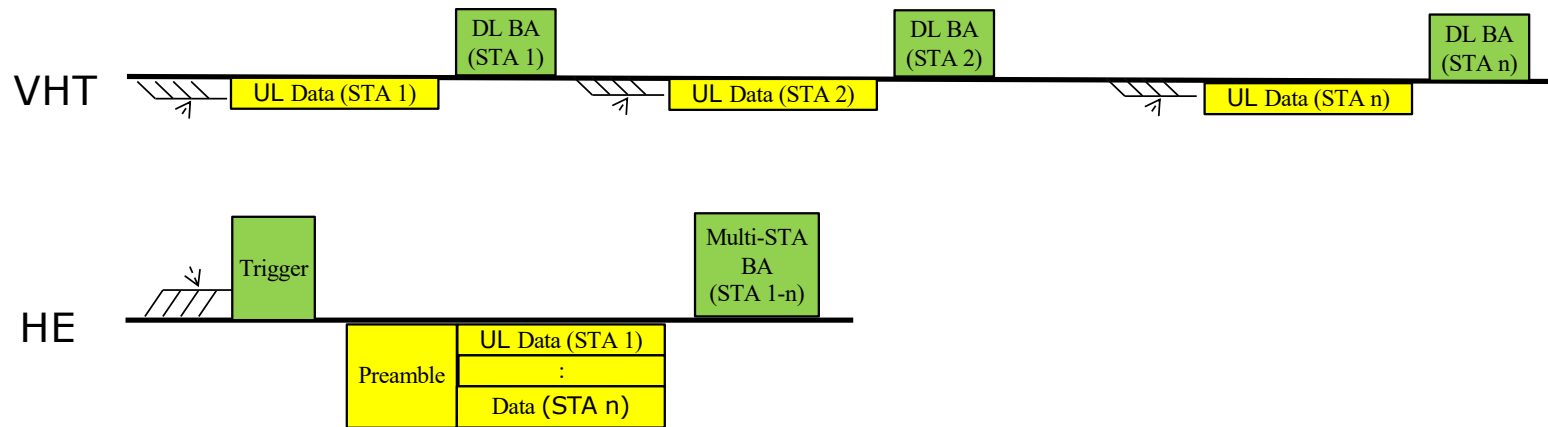


Modulation
(~25% gain)



+ 1024-QAM

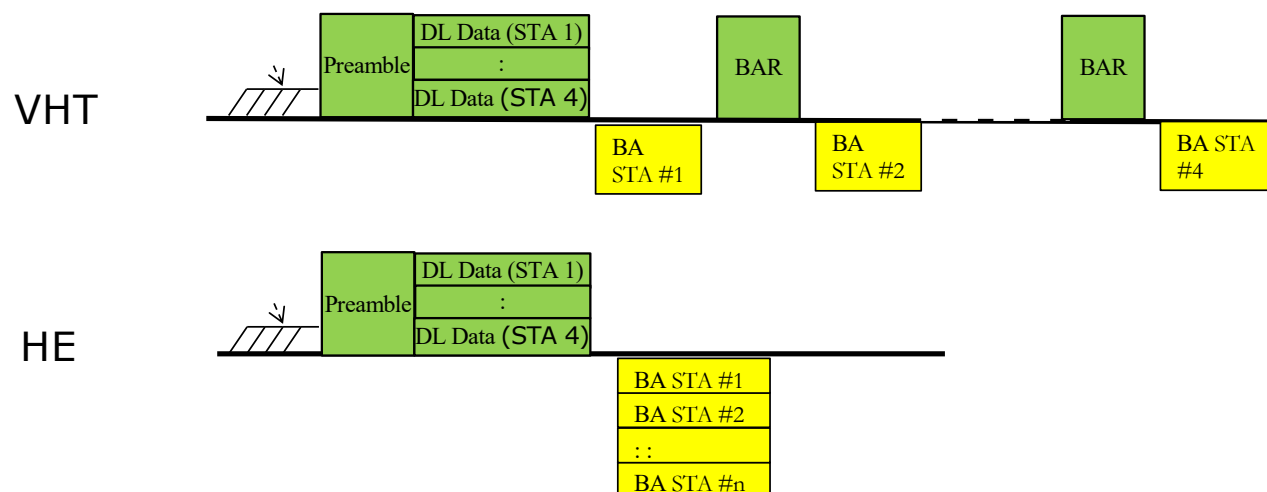
UL/DL multi-user links in 802.11ax will support more efficient UL data



- In a VHT UL sequence, STAs compete for medium access and send sequentially
- In an HE UL sequence, the AP triggers simultaneous transmissions in multiple STAs

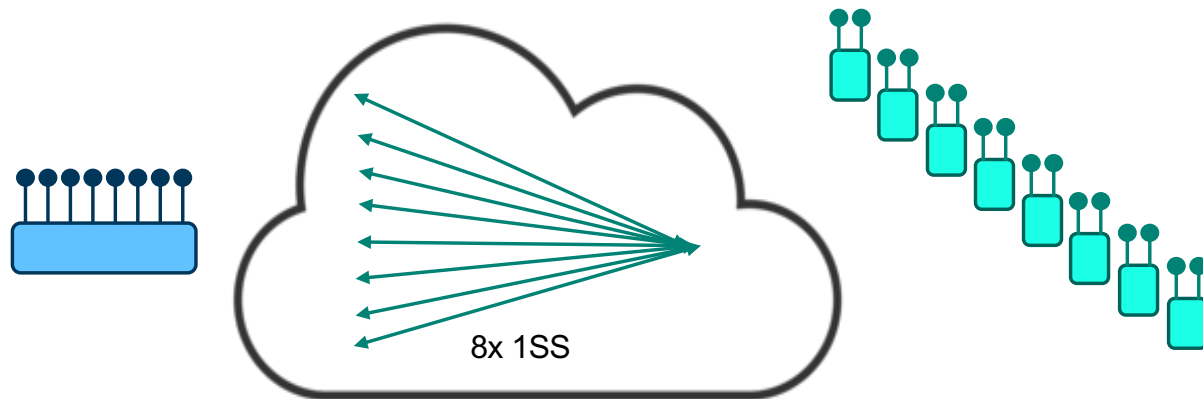
802.11ax Data exchange sequences: Multi-user downlink

- In a VHT DL MU sequence acknowledgements are serialized
- In an HE DL MU sequence acknowledgements are allocated UL resources and transmitted simultaneously



Uplink MU-MIMO

- UL MU-MIMO was initially considered in 802.11ac, but not included due to implementation concerns
- Sounding frames, data frames, etc can be grouped among multiple users to reduce overhead and increase uplink response time



Various features in 802.11ax will support improved outdoor operation

- Operates in higher delay spread channels than 802.11ac:
 - 802.11ac GI options: 0.4 μ s and 0.8 μ s
 - 802.11ax GI options: 0.8 μ s, 1.6 μ s and 3.2 μ s
 - GI overhead mitigated with longer OFDM symbol
- Preamble includes repeated L-SIG
- Extended range preamble includes repeated HE-SIG-A
- Dual carrier modulation improves robustness in Data field

802.11ax meets the MAC/PHY requirements for 5G Indoor Hotspot test Environment defined by IMT-2020

- Analysis and simulations confirm that performance of IEEE 802.11ax MAC/PHY meet or exceed 5G requirements for the 5G Indoor Hotspot use case
- Similar studies are underway for the Dense Urban test environment

	Metric	ITU-R Evaluation Method	Minimum Requirement	802.11ax Performance
1	Peak data rate	Analytical	DL/UL : 20/10 Gbps	DL/UL : 20.78 Gbps [Note 1]
2	Peak spectral efficiency	Analytical	DL/UL : 30/15 bits/s/Hz	DL/UL : 58.01 bits/s/Hz [Note 2]
3	User experienced data rate	Analytical for single band and single layer; Simulation for multi-layer	Not applicable for Indoor Hotspot	Not applicable
4	5 th percentile user spectral efficiency	Simulation	DL/UL : 0.3/0.21 bits/s/Hz	DL/UL : 0.45/0.52 bits/s/Hz [Note 3]
5	Average spectral efficiency	Simulation	DL/UL : 9/6.75 bits/s/Hz/TRxP	DL/UL : 9.82/13.7 bits/s/Hz/TRxP [Note 3]
6	Area traffic capacity	Analytical	DL : 10 Mbit/s/m ²	Required DL bandwidth = 170 MHz with 3 TRxP/site. [Note 4]
7	Mobility	Simulation	UL : 1.5 bits/s/Hz	UL : 9.4 bits/s/Hz
8	Bandwidth	Inspection	100 MHz, scalable	20/40/80/80+80/160 MHz
9	User plane latency	Analytical	DL/UL : 4 ms	DL/UL : 80 us [Note 5]

802.11ay is defining next generation 60 GHz: increased throughput and range

- 20Gbps+ rates are defined
- License- Exempt bands above 45Gbps
- Completion in 2020; First chipsets announced

Use Cases:

- Ultra-Short Range
- 8K UHD - Smart Home
- AR/VR and wearables
- Data Center Inter Rack connectivity
- Video / Mass-Data distribution
- Mobile Offloading and MBO
- Mobile Fronthauling
- Wireless Backhauling (w. multi-hop)
- Office Docking
- Fixed Wireless

Key additions :

- SU/ MU MIMO, up to 8 spatial streams
- Channel bonding
- Channel aggregation
- Non-uniform constellation modulation
- Advanced power saving features

60 GHz Fixed Wireless Use Case: Affordable 5G Performance

“the 14 GHz of contiguous spectrum in the band offers more bandwidth than any other licensed or unlicensed mmWave band. Further, the 60 GHz band has chipsets and technology currently available on the commercial market.”

“In the U.S., unlicensed mmWave frequencies available for 5G primarily cover the band from 57 – 71 GHz, called the V-Band, or 60 GHz band. This band offers 14 GHz of contiguous spectrum, which is more than all other licensed and unlicensed bands combined⁷. **This makes the 60 GHz band an excellent alternative to licensed mmWave frequencies for smaller providers, as it can be used to deliver 5G performance for the minimal cost of available 60 GHz infrastructure products.**

<https://go.siklu.com/hubfs/Content/White%20Papers/Maravedis%20Industry%20Overview:%205G%20Fixed%20Wireless%20Gigabit%20Services%20Today.pdf>

<https://www.fiercewireless.com/wireless/60-ghz-band-particularly-appealing-for-fixed-wireless-report>

60 GHz Mesh Backhaul Wireless Use Case: Deploying Today

*“Leading Wi-Fi and wireless network solution vendor Cambium Networks announced today that they will be incorporating Facebook’s Terragraph technology into a new series of Cambium Networks **60 GHz radio products** called **cnWave™**. The news comes as Terragraph appears to be ramping up go-to-market activities with trials underway in Hungary and most recently in Malaysia.”*

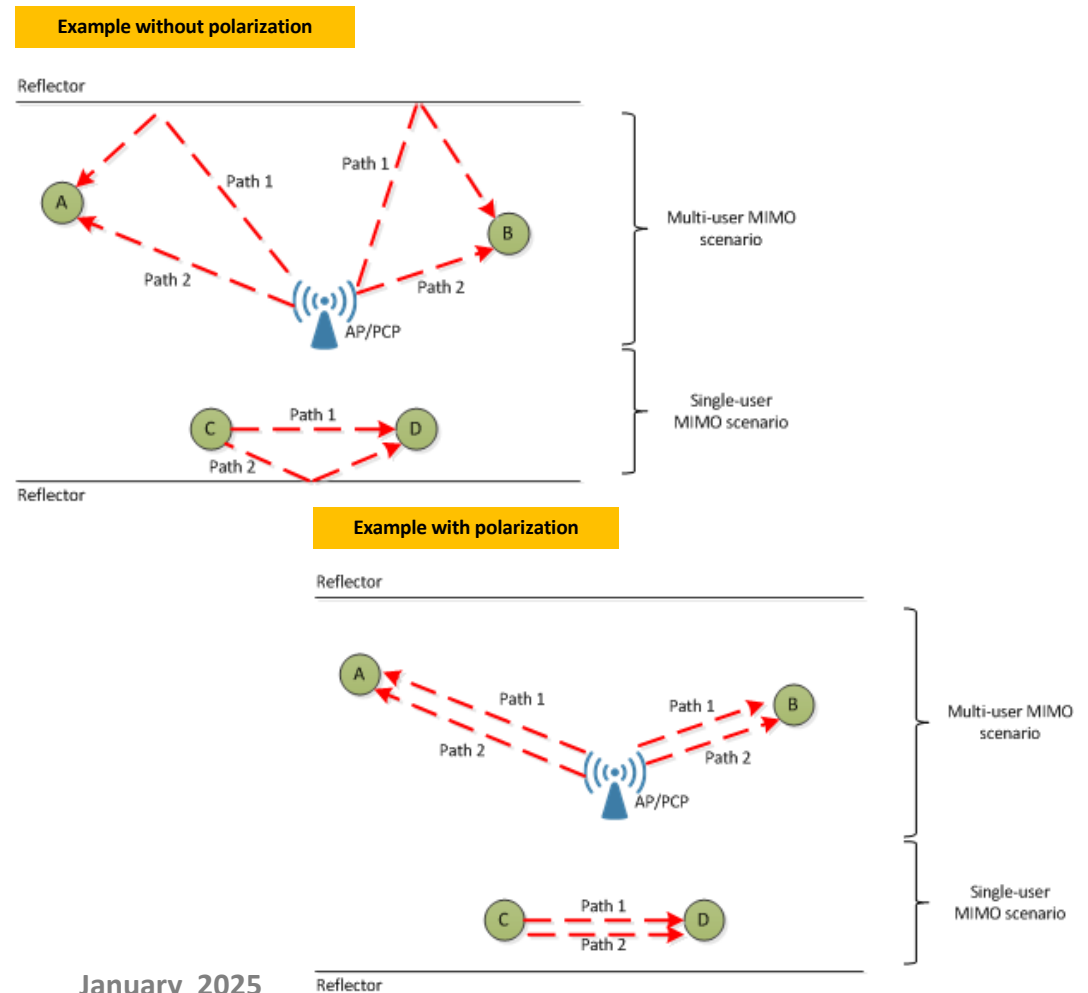
“Terragraph is essentially a 60 GHz-based meshed (or multi-hop, multi-point) backhaul radio system for deployment at street level in cities.”

<https://wifinowevents.com/news-and-blog/cambium-networks-to-incorporate-facebook-terragraph-tech-into-new-60-ghz-products/>

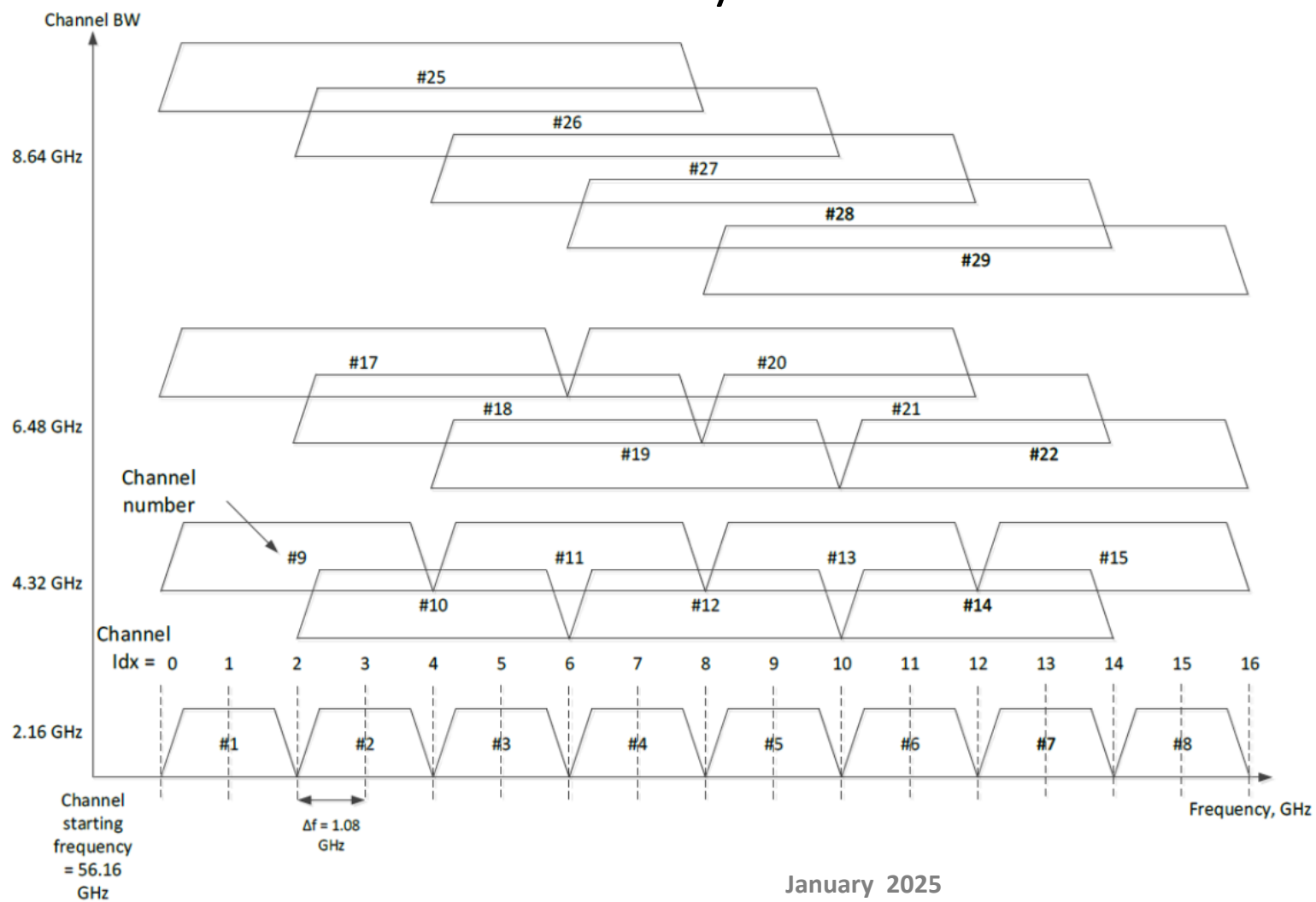


802.11ay builds on 802.11ad with MIMO and channel bonding features

- Channel bonding and aggregation requires new:
 - Channelization
 - Packet format
 - Channel access mechanisms
- Single User and downlink MU MIMO
 - Distribute capacity across users
 - Unique requirements given directionality
 - Exploit antenna polarization
 - Changes to the beamforming protocol



802.11ay defined channelization



802.11az Next Generation Positioning

The P802.11az project Next Generation Positioning extends accurate IEEE Std 802.11-2016 Fine Timing Measurement capabilities

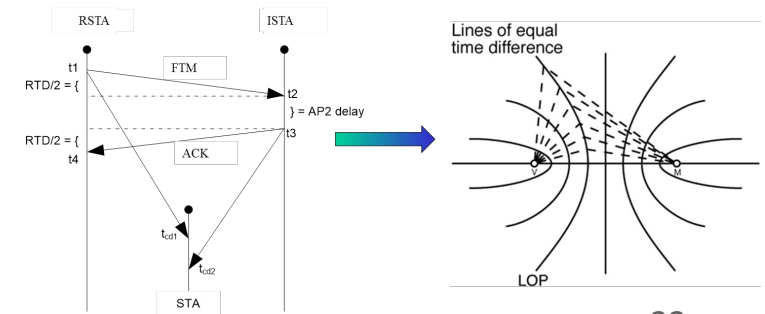
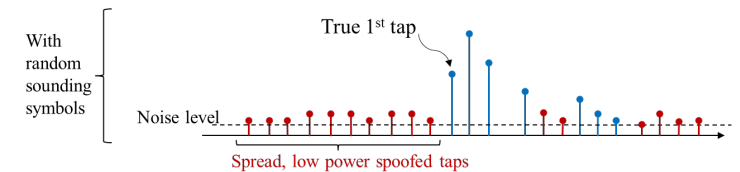
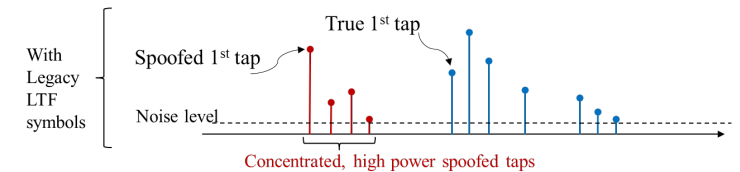
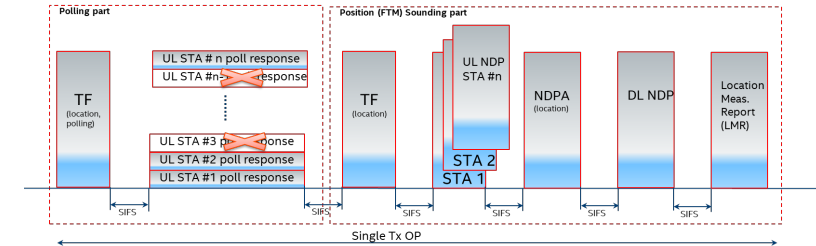
- Accurate indoor Navigation (sub 1m and into the $<0.1\text{m}$ domain).
- Enables self-locating networks for easy, fast and cost efficient WLAN deployment for navigation and 6GHz AFC operation.
- Secured (authenticated and private) positioning – open my car with my smartphone, position aware services (money withdrawal).
- Unlock computer with a wearable device, adapt TV content to audience presence.
- Location based link adaptation for home usages (connect to best AP).
- Navigate in extremely dense environments (stadium/airport scenarios).



January 2025

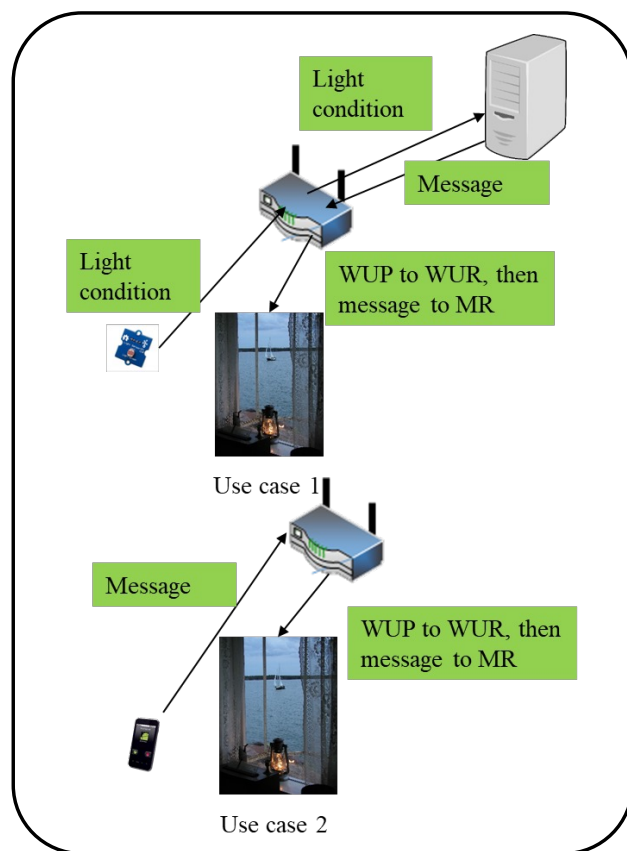
802.11az Key Radio and Positioning Techniques

- Medium efficient operation via dynamic (demand dependent) measurement rate
- Adaptation to next generation mainstream 802.11ax Trigger Based Operation (MIMO, Trigger Frame, NDP frame)
- Authenticity and privacy and anti-spoofing mechanism via PMF in the unassociated mode and PHY level randomized measurement sequences (HE LTF sequences protection)
- Improved accuracy via MIMO and larger BW available in the <7Ghz band for 11ax
- MIMO enablement for measurement for improved accuracy especially for NLOS or NNLOS conditions
- Passive location with fixed overhead independent of number of users

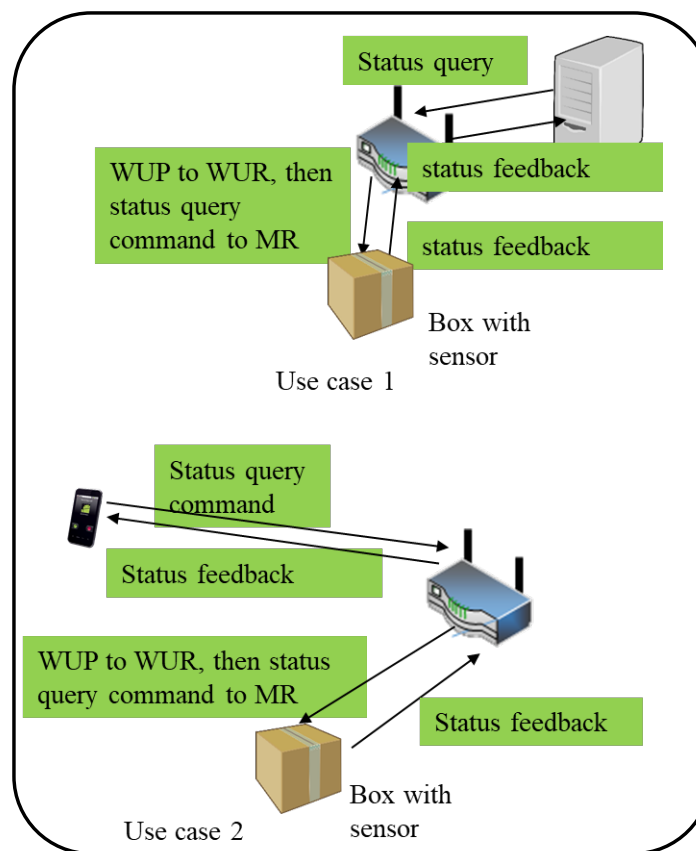


802.11ba Wake-up Radio Main Use Cases

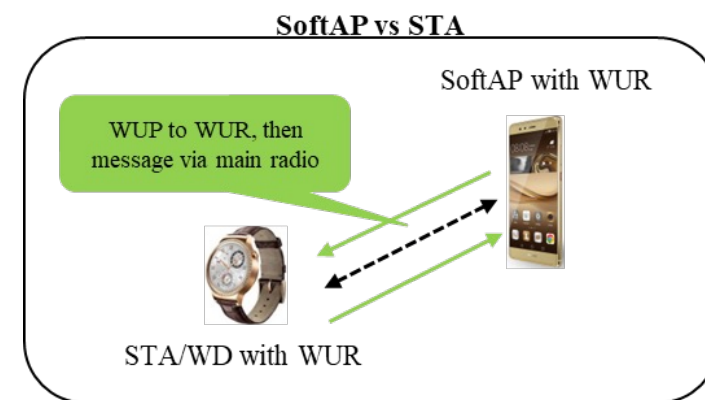
1. Smart Home



2. Warehouse

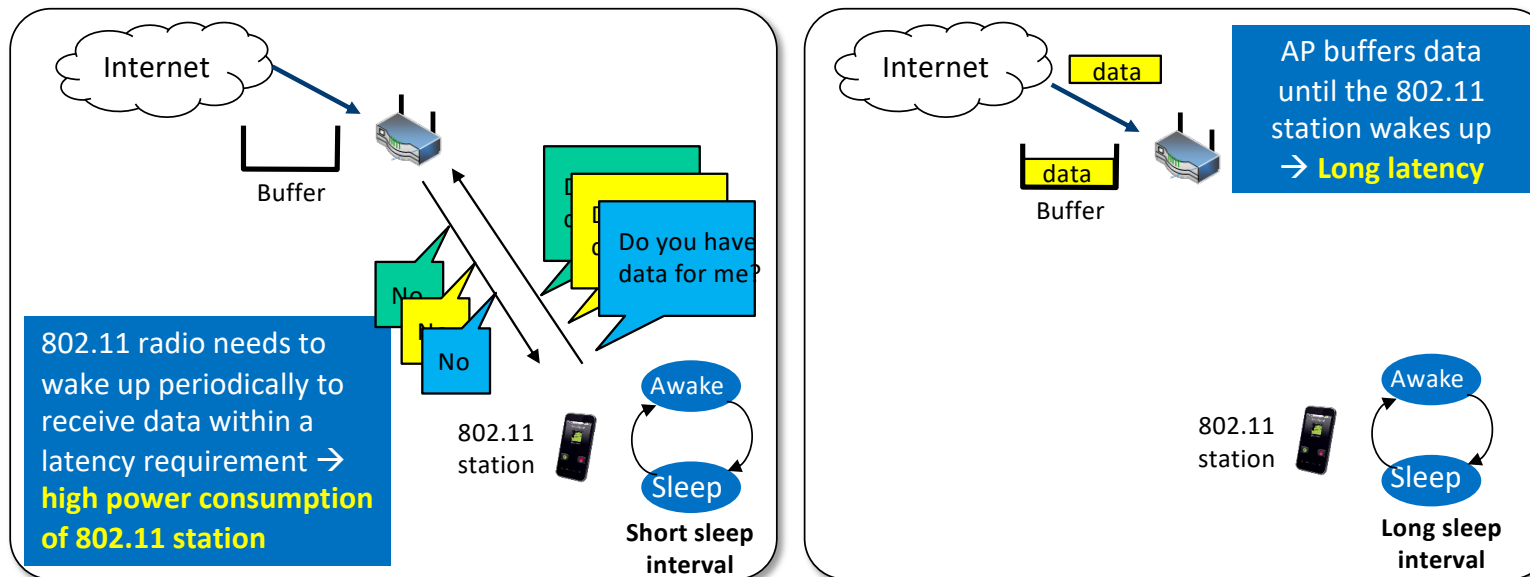


3. Wearables



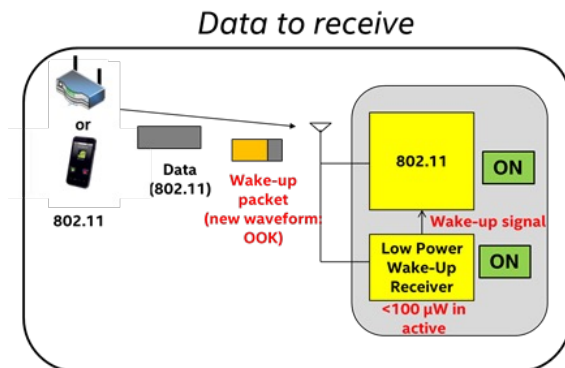
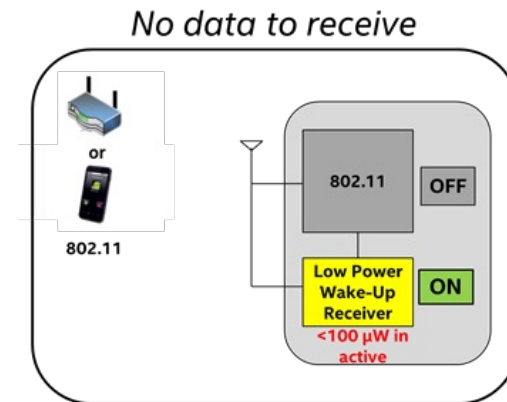
WUP: wake-up packet
WUR: wake-up receiver
MR: main radio

802.11ba improves energy efficiency of stations and maintains low latency

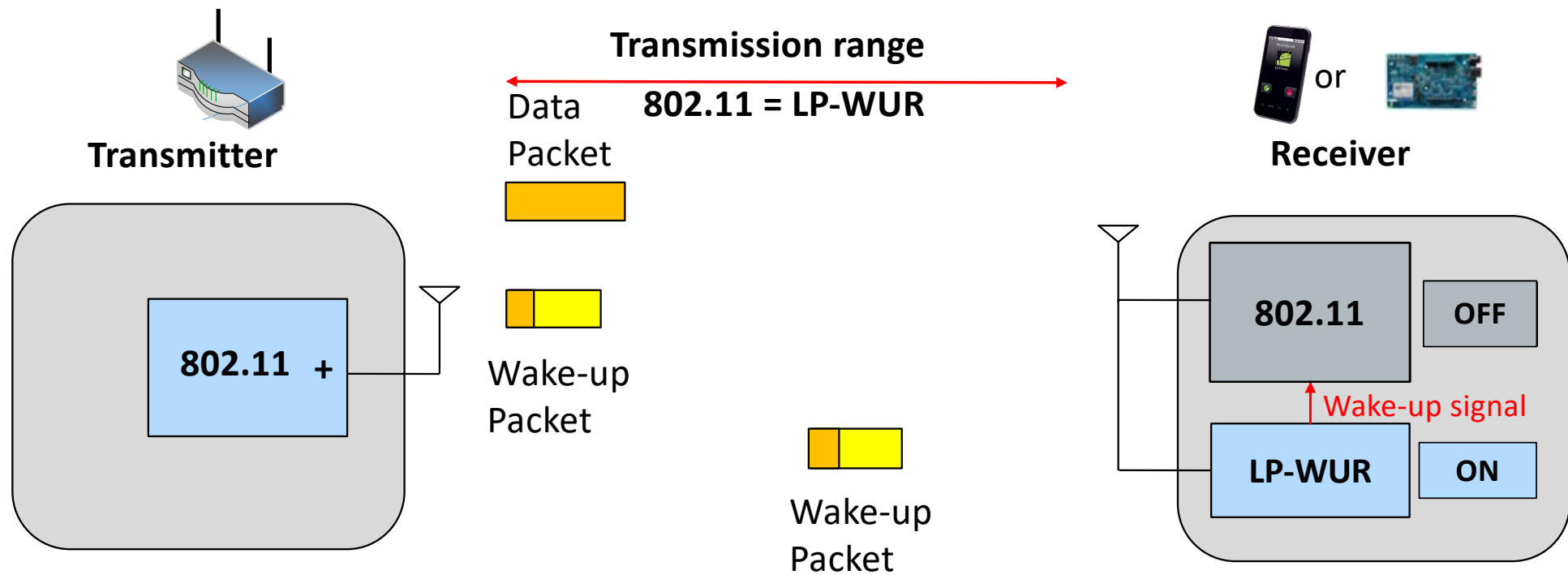


802.11ba Low-power Wake-up Receiver (LP-WUR) as Companion Radio for 802.11

- Comm. Subsystem = Main radio (802.11) + LP-WUR
 - **Main radio (802.11): for user data transmission and reception**
 - Main radio is off unless there is something to transmit
 - LP-WUR wakes up the main radio when there is a packet to receive
 - User data is transmitted and received by the main radio
 - **LP-WUR: not for user data; serves as a simple “wake-up” receiver for the main radio**
 - LP-WUR is a simple receiver (doesn't have a transmitter)
 - Active while the main radio is off
 - Target power consumption < 1 mW in the active state
 - Simple modulation scheme such as On-Off-Keying (OOK)
 - Narrow bandwidth (e.g. < 5 MHz)
 - Target transmission range: LP-WUR = Today's 802.11



802.11ba Low Power Wake-up Radio Operation



802.11bb Light Communications

- Light Communications (LC)
 - Ability to re-use existing HT, VHT and HE PHY/MAC

Use Cases:

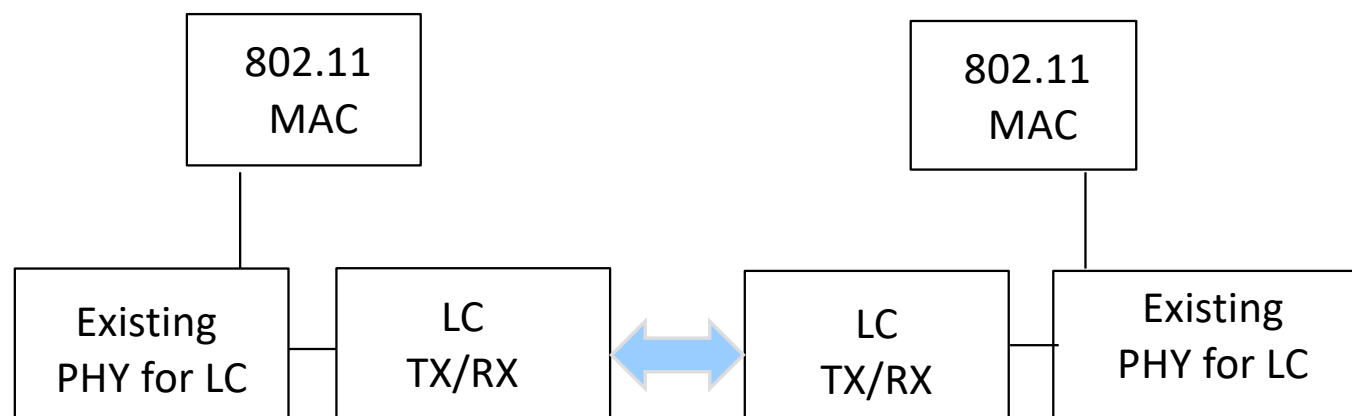
- Industrial wireless applications
- Medical environments
- Enterprise
- Home
- Backhaul
- Vehicle to Vehicle Communication
- Underwater Communication
- Gas Pipeline Communication

Key additions :

- Uplink and downlink operations in 800 nm to 1000 nm band
- Minimum single-link throughput of 10 Mb/s
- Interoperability among solid state light sources with different modulation bandwidths.

802.11bb transports defined 5/6GHz MAC/PHY across the light channel

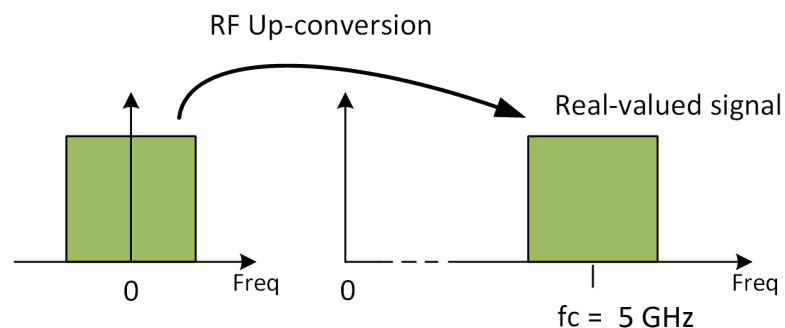
5/6 GHz 802.11 MAC and PHY are used as defined today



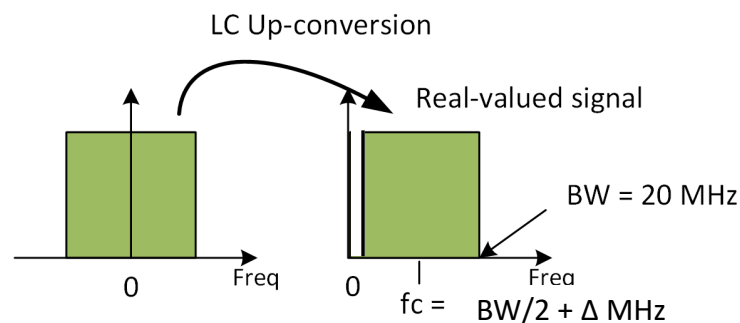
A Physical Layer Protocol Data Unit (PPDU) is converted to light using intensity modulation. At the receiver, the optical signal is converted to an electrical signal. LC systems transmit on wavelengths within the 800nm to 1000 nm band.

802.11bb uses light spectrum and existing WLAN capabilities

- RF frontend up-converts baseband signals onto e.g. $f_c = 5$ GHz.



- LC frontend up-converts baseband onto low IF e.g. $f_c = BW/2 + \Delta$.

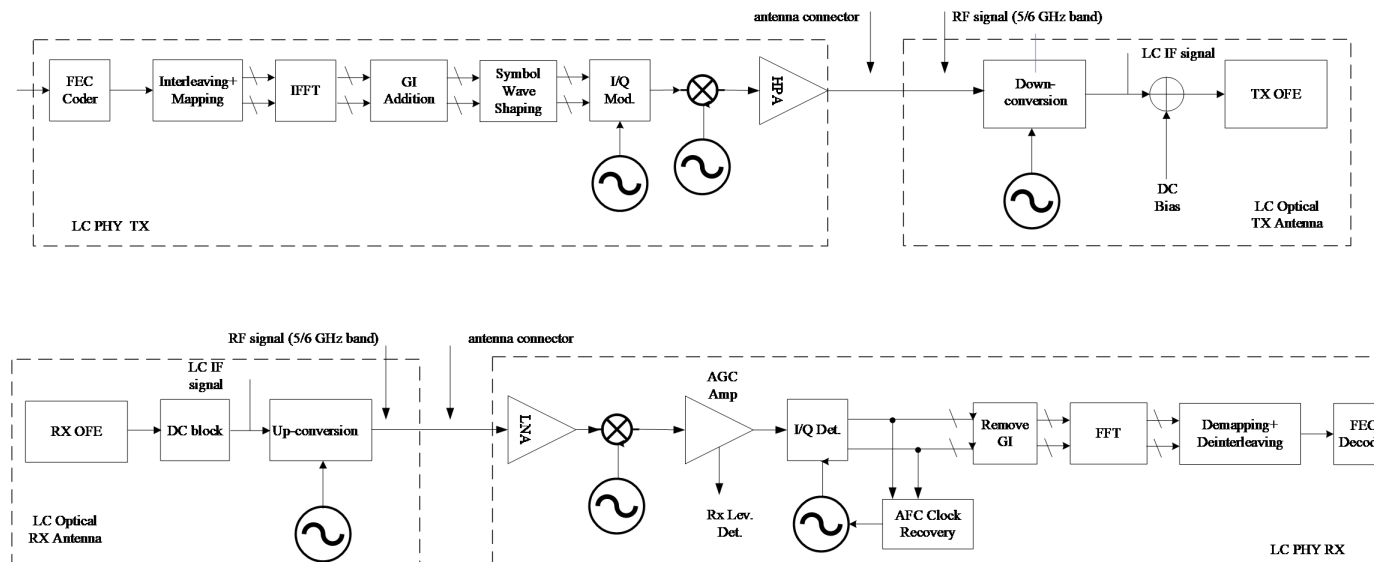


– Δ is specified in the existing draft of TGbb for various channel bandwidths.

- In this manner, LC re-uses the existing HT, VHT and HE PHY/MAC specifications.

802.11bb uses light spectrum and existing 802.11 technical capabilities

- Example implementation of P802.11bb using up- and down-conversion of existing 5 GHz or 6 GHz chipsets



- Interfacing the LC PHY to light emitter and receiver.

802.11bb usage model 1: Industrial wireless

Pre-Conditions

Devices may experience unstable radio frequency (RF) connection due to Electro-Magnetic Interference (EMI) in factories. LC is deployed to provide reliable wireless connectivity for industrial wireless networks.

Environment

All communications are within a large metal building, industrial or automated work cell. The area of these environments range from tens to thousands of square meters, equipped with industrial robot and other equipment. The environment has high levels of EMI. Lighting level of 150 lux is recommended (1500 lux for dedicated work).

Applications

Ultra-high-definition (UHD) video streaming for surveillance or production monitoring (quality control) applications, for video collaboration for team, customer, and supplier meetings. Lightly compressed Video: ~ 1Gbps, delay < 5 ms, 1×10^{-8} PER, 99.9% reliability. Fully connected factory—for real-time communications, application execution, and remote access. Distance between LC APs ranges from 2~20 meters.

Traffic Conditions

Both uplink and downlink traffic is using LC. High levels of OBSS interference between LC access points (APs) expected due to very high density deployment. Potential non-LC interference from surrounding environments such as artificial-light. Multiple LC modules are deployed on the robot/equipment and on the ceiling/walls to provide multiple light links for a robust connectivity in case a single line-of-sight (LOS) link is blocked.

Use Case

An industrial robot is powered on and ready for operation. Operating instructions are transmitted to the robot via LC. The robot is working (e.g., movement) according to the instructions and provides real-time feedback information and/or video monitoring data for quality control to control center also via LC. Upon command, the robot finishes the task and is ready for the next one.

802.11bb usage model 2: Wireless access in medical environments

Pre-Conditions

IEC 60601-1-2 standard recommends the minimum separation distance between medical electrical (ME) equipment and RF wireless communications equipment (e.g., wireless local area network (WLAN)) be 30 cm to avoid performance degradation of the ME equipment. LC is deployed to ensure the performance of all ME equipment.

Environment

The size of a operating theater and MRI room ranges from 30~60 m². Multiple LC-APs are deployed on the ceiling to provide specialized illumination. The central illuminance of the operating light: 160k and 40k lux. The size of a four beds ward is about 60 m², light level: 300 lux on the bed and >100 lux between the beds and in the central area.

Applications

LC-WLAN is used to allow wireless data exchange in medical environments with ME equipment or system. Medical multimedia and diagnostic information can be transmitted to provide telemedicine services; ME equipment can also be wirelessly controlled via LC. Provide Intranet/ Internet access, audio or video call for doctors, nurses and patients using LC-based devices.

Traffic Conditions

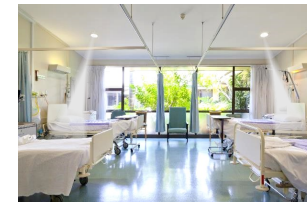
No interference caused by RF radiation.
Both uplink and downlink traffic is ~~are~~ using LC.
High Quality of Service (QoS) and high reliability are required.
Potential non-LC interference from surrounding environments such as artificial-light.

Use Case

Doctors enter an operating theater, turn on the LC enabled LED lights and ME equipment. Doctors can interact with the remote doctors and share information using LC. ME equipment connectivity is also supported by LC. Doctors finish the treatment, then turn off the lights and medical equipment. A patient is monitored by ME equipment which communicate with the nurses/doctors in control room via LC.



Operating theater



Hospital ward

802.11bc is defining Enhanced Broadcast Services

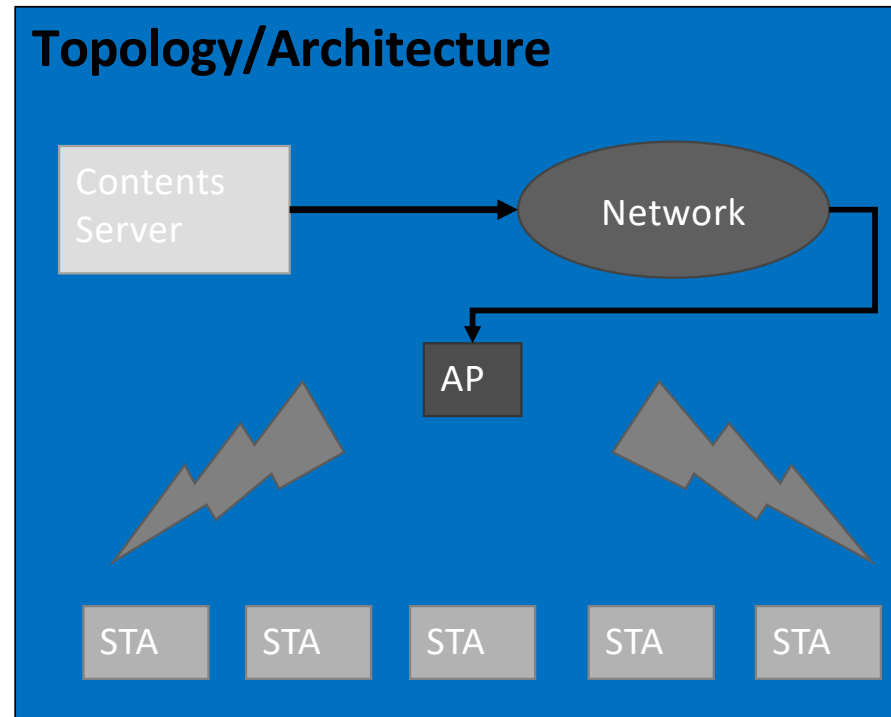
- Enhanced Broadcast Services (eBCS) defines broadcast service enhancements within an 802.11-based network.
- Client end devices broadcast information to an AP, e.g. in an IoT environment, to other STAs so that any of the receiving APs act as an access node to the Internet.

802.11bc Broadcast Downlink use case description

Broadcast Downlink

Provides enhanced Broadcast Services (eBCS) of data (e.g. videos) to a large number of densely located STAs.

These STAs may be associated, or un-associated with the AP or may be low-cost STAs that are receive only.

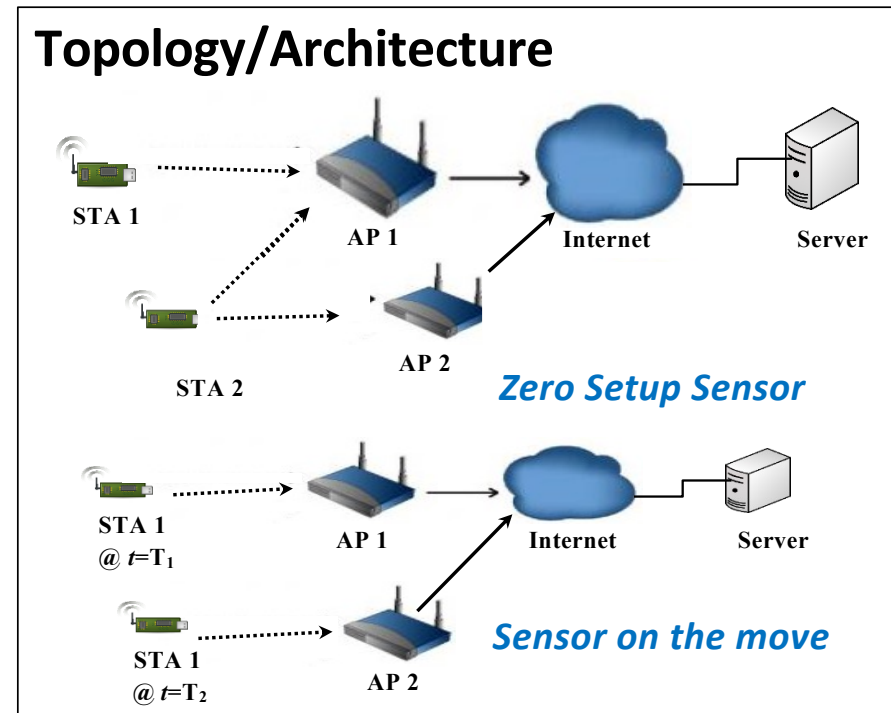


802.11bc Broadcast Uplink use case description

Broadcast Uplink

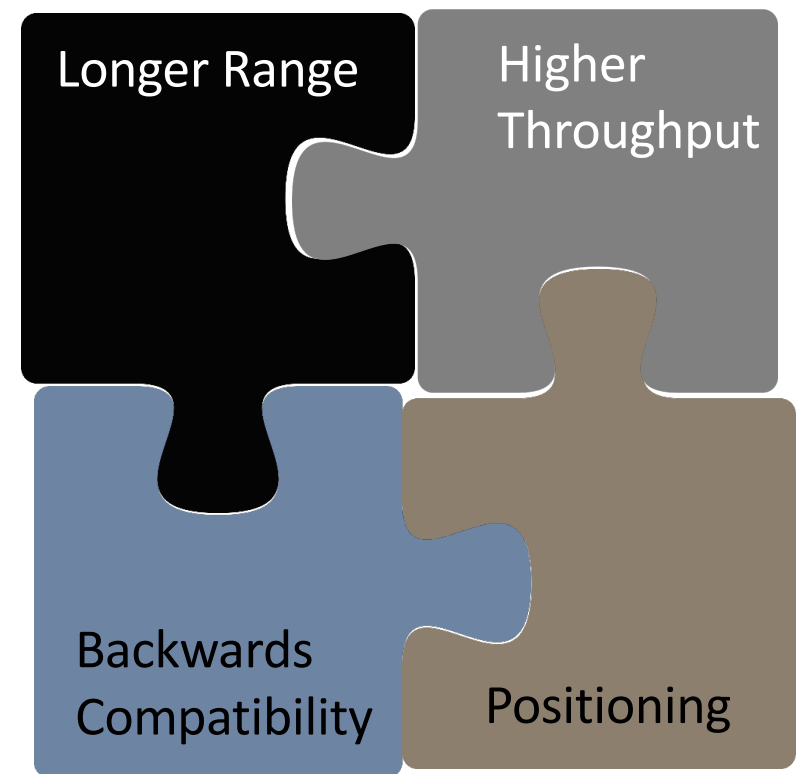
Pre-configured devices (e.g. IoT) automatically connect to the end server through APs with zero setup action required.

Alternatively, low power IoT devices that are in motion, report to their servers through APs without scanning and associating



802.11bd defines an evolution of 802.11p for Vehicle to Anything (V2X)

- 802.11p is largely based on 802.11a.
- 802.11bd defines MAC/PHY enhancements from 802.11n, ac, ax, to provide a backwards compatible next generation V2X protocol.
- **Higher Throughput**
 - OFDM frame design
 - Higher MCS, LDPC coding
 - Packet aggregation
- **Longer Range**
 - Mid-amble design
 - Repeated transmission mechanism
 - More robust channel coding
- **Support for Positioning**
- **Backward Compatibility**
 - Backward compatible frame format design, Version indication



802.11bd: Next Generation V2X Use Cases

5.9 GHz band mainly, and optionally 60 GHz;

Completion in 2022/2023

http://www.ieee802.org/11/Reports/tgbd_update.htm

V2X Use Cases:

- Support all defined DSRC/802.11p use cases, including Basic safety message (safety, range, backward compatibility, fairness)
- Sensor sharing (throughput)
- Multi-channel operation (safety channel + other channels)
- Infrastructure applications (throughput)
- Vehicular positioning & location (LoS and NLoS positioning accuracy)
- Automated driving assistance (safety, throughput)
- Aerial vehicle IT application (video)
- Train to train (high speed)
- Vehicle to train (high speed, long range)

Key additions :

- Higher throughput (2x) than 802.11p
- Longer range (3dB lower sensitivity level)
- Support for positioning
- Backward compatibility with 11p

802.11be Extremely High Throughput amendment builds on 802.11ax, including 6GHz support

Extremely High Throughput (EHT)

Operation in 2.4 GHz, 5 GHz, and 6 GHz bands

Higher throughput – Project goal of at least 30 Gbps; expect 40+Gbps with 320MHz channels, 4096 QAM

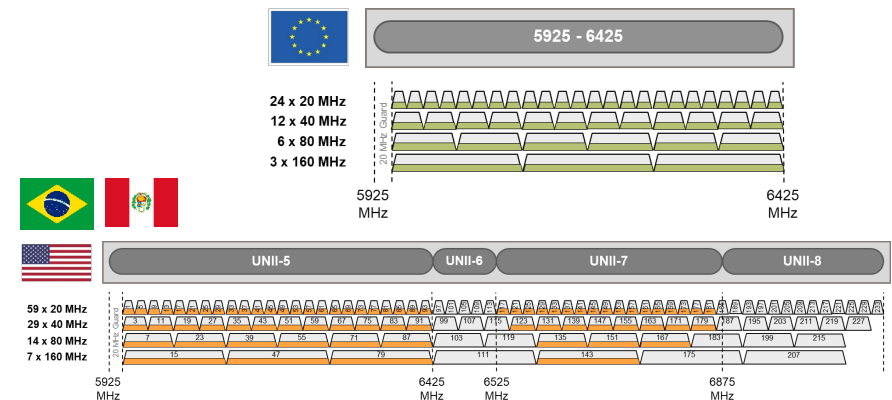
Support for low latency communications

Continued improvements in spectral efficiency

Targeted completion in 2024

Use Cases:

- Home, enterprise, industrial, IoT
- Outdoor
- AR/VR, wireless gaming
- 4K and 8K video streaming
- Remote office
- Cloud/edge computing
- Video calling and conferencing

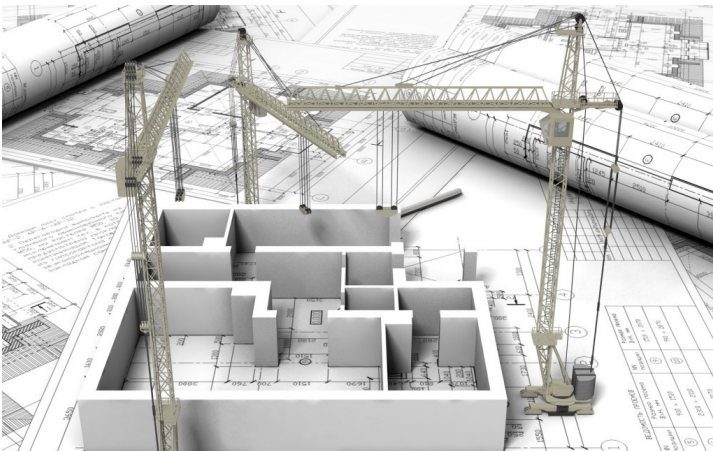


802.11be features under consideration

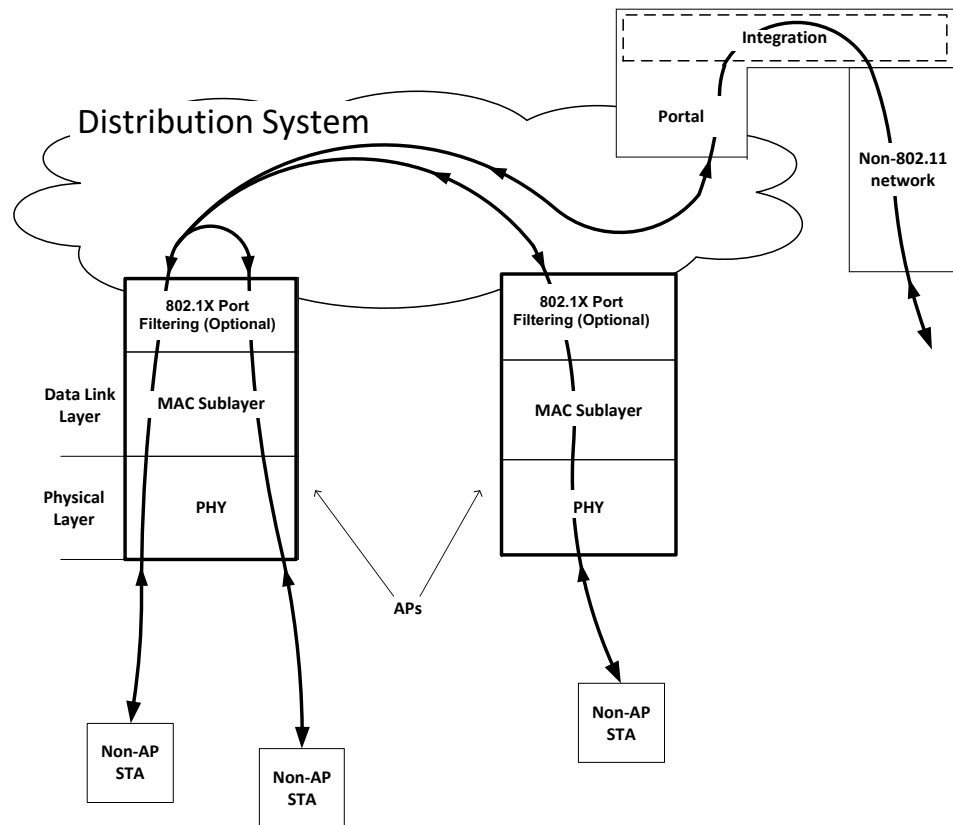
- Support of 320 MHz bandwidth channels, enabling extremely high throughput use cases
- More efficient utilization of non-contiguous spectrum
- Multi-band/multi-channel aggregation and operation
- MIMO protocol enhancements, enhanced sounding protocol
- Multi-AP Coordination (e.g. coordinated and joint transmission)
- Refinements of 802.11ax features: Multiple Resource Unit operation
- Target Wait Time (TWT) enhancements
- GCMP-256 support
- Adaptation to regulatory rules specific to 6 GHz spectrum
- NSEP priority access operation

The ARChitecture SC (ARC) meets on an ongoing basis

- It does not develop a standard or amendment
- Discussion of 802.11 architecture issues for other ongoing amendments
- Topics for discussion include:
 - Architectural clauses and models in the standard, as questions, or needs for update or clarification arise.
 - Relationships with outside groups on 802.11 architectural topics, or topics that don't fit elsewhere, such as IETF, 802 O&A, and 802.1.



ARC technical considerations



- Models for STA architecture and related concepts, and overall system architecture, included in the Standard in clauses 4 and 5, generally.
- Evolution of the models, either to consider amendments to the Standard, or as clarification is needed
- Define how 802.11 technologies fit into 802, 802.1 and IETF use cases.

Thank You

Questions